Explainability of a Heterogeneous Temporal Graph Neural Network for Forex Allocation

AI Ethics and Explainability

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Abstract

Foreign exchange allocation is a high-stakes prediction setting because the model output is not an isolated forecast but a capital allocation across relative macro assets. This report studies the explainability of a Heterogeneous Temporal Graph Neural Network (HTGNN) that allocates across USD, EUR, JPY, GBP, CNY, CAD, AUD, and CHF using a portfolio-signal node and macro-financial context nodes such as FX futures, currency pairs, rates, commodities, and equity risk. The goal is not only to report backtest performance, but to assess whether the model's decisions can be related to interpretable market channels. The analysis combines three families of XAI methods: a surrogate SHAP analysis restricted to the portfolio-signal node, gradient-based Integrated Gradients over the full heterogeneous input graph, and perturbation-based temporal occlusion plus deletion/insertion faithfulness checks. The results show a profitable but diversified HTGNN backtest and explanations that consistently point to FX derivatives, commodity-linked information, recent temporal windows, and the portfolio signal as relevant decision drivers. The report also discusses limitations: short evaluation horizon, low-fidelity linear SHAP surrogate, baseline sensitivity, turnover risk, and the need for explanation-stability monitoring before any domain deployment.

1 Introduction and Motivation

Foreign exchange (Forex or FX) is inherently relative: each currency is priced against another numeraire. This project fixes USD as numeraire and allocates across USD, EUR, JPY, GBP, CNY, CAD, AUD, CHF. The prediction target is therefore not a price, but a long-only portfolio $\boldsymbol{w}_t \in \Delta^N$, with $\Delta^N = \{\boldsymbol{w} \in \mathbb{R}^N_+ : \mathbf{1}^T \boldsymbol{w} = 1\}$.

Forex is an interesting domain because currencies are macro assets. Their returns are linked to monetary policy, inflation, external balances, commodity exposure, safe-haven flows, and global risk appetite. These drivers are not independent scalar features; they are naturally organised into market blocks. The HTGNN hypothesis is that a typed temporal graph can use those blocks more appropriately than a signal-only architecture.

The ethical and explanatory motivation is just as important as the predictive one. A currency allocation model affects capital allocation and risk exposure. Relevant stakeholders include portfolio managers, model validators, risk officers, clients, and counterparties exposed to the resulting trades. For these stakeholders, a high backtest return is not enough: the model should provide evidence that its allocations are driven by plausible market channels rather than by leakage, unstable noise, or accidental correlations. The report therefore focuses on the question: which nodes, windows, and input currencies appear to drive the HTGNN's allocation decisions?

2 Data and Model

The repository uses daily adjusted Yahoo Finance price series, downloaded with `yfinance` [1], from 2017-01-01 to 2025-12-31. Prices are converted to log returns, short gaps are forward-filled with a three-day limit, and instruments with excessive missingness are removed. The supervised tensors use a 20-day lookback and chronological splits: 2018-01-01 to 2024-01-01 for training, 2024-01-01 to 2025-01-01 for validation, and 2025-01-01 to 2026-01-01 for test. The models are implemented in PyTorch [2].

For every non-USD currency, the Yahoo Finance quote `USDXXX=X` is interpreted as units of currency XXX per one USD. Hence the USD-denominated return of holding currency XXX has the opposite sign of the quoted pair return:

$$r^{\mathrm{USD}}_{t,\mathrm{XXX}} = -\Big(\log P^{\mathrm{USD/XXX}}_t - \log P^{\mathrm{USD/XXX}}_{t-1}\Big), \quad r^{\mathrm{USD}}_{t,\mathrm{USD}} = 0. \tag{1}$$

The final preprocessing centres the eight currency returns cross-sectionally before window construction. This matters for explainability: without centring, the raw USD channel would always be zero; after centring, USD represents the negative average movement of the basket and can receive non-zero attribution.

Training uses a mean-variance teacher inspired by portfolio optimisation [3]. Let $\boldsymbol{r}_t \in \mathbb{R}^N$ be the USD-valued currency return vector and R_t the recent lookback window. The target mean proxy is the next realised cross-section centred around its own average, $\tilde{\boldsymbol{\mu}}_{t+1} = \boldsymbol{r}_{t+1} - \frac{1}{N}\mathbf{1}\mathbf{1}^T \boldsymbol{r}_{t+1}$. The covariance $\hat{\Sigma}_t$ is estimated from the lookback window with a ridge term, and the ex-post teacher allocation is

$$\boldsymbol{w}^*_{t+1} = \arg\max_{\boldsymbol{w} \in \Delta^N} \boldsymbol{w}^T \tilde{\boldsymbol{\mu}}_{t+1} - \frac{\gamma}{2} \boldsymbol{w}^T \hat{\Sigma}_t \boldsymbol{w}. \tag{2}$$

The next return is used only to build the supervised teacher during training. At inference, the model receives past node windows and outputs an allocation.

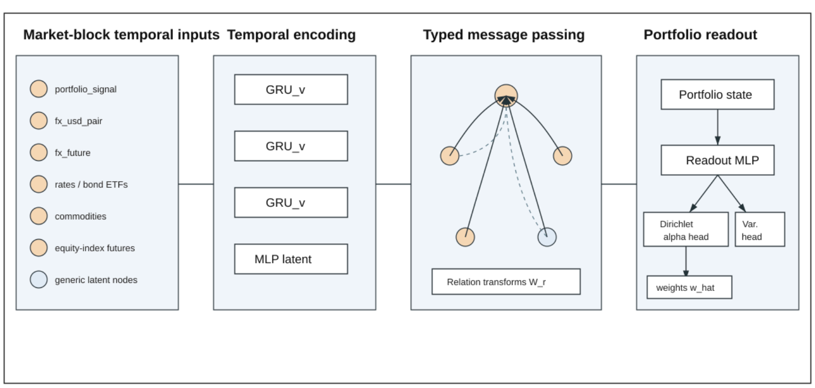

Figure 1: Implemented HTGNN architecture. Each market block is encoded as a temporal node, relation-specific message passing is applied over a portfolio-centred heterogeneous graph, and the final allocation is read from the portfolio node.

For each observed node v, the input is a window $X_{v,t-L+1:t} \in \mathbb{R}^{L \times F_v}$. Each node has its own GRU input dimension, but all nodes are mapped to a shared hidden size H: $\boldsymbol{h}_v^0 = \mathrm{GRU}_{v(X_{v,t-L+1:t})} \in \mathbb{R}^H$. After temporal encoding, the HTGNN applies relation-typed message passing, combining the motivation of temporal graph modelling [4] with relational and heterogeneous graph models [5], [6]:

$$\boldsymbol{m}_v^\ell = \frac{1}{|}\mathcal{N}(v)| \sum_{(u,v,r)\in\mathcal{E}} W_r^\ell \boldsymbol{h}_u^\ell, \tag{3}$$

$$\boldsymbol{h}_v^{\ell+1} = \mathrm{LayerNorm}\Big(\boldsymbol{h}_v^\ell + \mathrm{MLP}_{\ell([\boldsymbol{h}_v^\ell;\boldsymbol{m}_v^\ell])}\Big). \tag{4}$$

The selected HTGNN uses hidden dimension 96, one GRU layer, relation-specific message passing, GELU activations, dropout 0.10, batch size 64, and a sliced Wasserstein allocation loss. The final `portfolio_signal` state is mapped to Dirichlet concentrations, $\boldsymbol{\alpha}_t = \mathrm{softplus}(\boldsymbol{a}_t) + \varepsilon$, and the predicted allocation is $\hat{\boldsymbol{w}}_t = \frac{\boldsymbol{\alpha}_t}{\mathbf{1}^T \boldsymbol{\alpha}_t}$.

3 XAI Methodology

The XAI pipeline is organised into three complementary classes: surrogate/approximation methods, gradient-based attribution methods, and perturbation-based methods. This classification is useful because each class answers a different question. Surrogates ask whether a simpler model can approximate part of the HTGNN. Gradients ask how a local output changes along an interpolation path. Perturbations ask what happens to realised performance when information is removed.

3.1 Surrogate / approximation method: portfolio-signal SHAP

SHAP values are based on additive feature attributions [7]. A full SHAP analysis over the HTGNN input would be too large: the model receives many nodes, each with a 20-day window and a different number of symbols, and the output has eight currency allocations. Instead, the implemented SHAP experiment focuses on the `portfolio_signal` node. This is not the whole model, but it is the most directly interpretable allocation-history input.

The full 20-day `portfolio_signal` window is flattened into 160 features: 20 lags times 8 currencies. For each output currency, a standardised ridge surrogate is fitted to the HTGNN output. Linear SHAP-style contributions are then computed as

$$\varphi_{j(x)} = \beta_j \big(z_j - \overline{z}_j\big), \tag{5}$$

where z_j is the standardised surrogate feature and β_j is the fitted coefficient. Contributions are summed across the 20 lags for each input currency, producing an 8-by-8 grid. The absolute SHAP grid is used to rank importance. The signed grid is also saved because it can reveal net directional tendencies, but signed means can be close to zero when positive and negative effects cancel. In this report, "mean" means the predicted mean allocation weight, not expected currency return.

3.2 Gradient-based attribution: Integrated Gradients

Integrated Gradients (IG) attributes a differentiable output to input features by integrating gradients along a path from a baseline x' to the observed input x [8]:

$$\mathrm{IG}_{i(x)} = (x_i - x_i') \int_0^1 \frac{\partial F(x' + \alpha(x - x'))}{\partial x_i} \, \mathrm{d}\alpha. \tag{6}$$

For this model, IG is computed over the observed heterogeneous nodes. The baseline is the mean tensor of each node over a background subset, not a zero tensor. This choice is important: after standardisation, zero may be meaningful for some channels, but replacing all nodes by zero is less faithful to the empirical graph distribution than using a background mean. The implementation uses 24 interpolation steps and aggregates absolute attributions by node. It also decomposes attribution inside each node to produce feature-level pie charts.

The explained score is aligned with the model's own allocation direction: for each local batch, the prediction is used as a direction vector and the scalar score is the inner product between predicted weights and that direction. This keeps IG focused on allocation behaviour rather than on a single arbitrary currency output.

3.3 Perturbation methods: occlusion and deletion/insertion

Occlusion methods replace part of the input and measure output or performance changes, an idea related to early perturbation-based explanation work [9]. Here the replacement value is again the background mean tensor per node. This is preferred to zero because masking should represent "typical missing information", not an artificial market state.

Temporal occlusion splits the 20-day lookback into five windows: 0–4, 4–8, 8–12, 12–16, and 16–20. In global temporal occlusion, the selected window is replaced by the background mean slice for every observed node. In node-window occlusion, the same operation is applied to one node at a time. The metric is the drop in performance ratio, where performance is annualized gross growth of the model strategy divided by annualized gross growth of an equal-weight portfolio.

Deletion/insertion is a faithfulness check inspired by perturbation tests such as deletion/insertion curves in vision explanations [10]. Nodes are ordered by explanation score. Deletion starts from the full input and progressively replaces the top nodes by their background baselines. Insertion starts from the complementary masked input and progressively restores the same nodes. Two views are reported: one relative to the equal-weight standard portfolio and one relative to the full unperturbed model. The report uses the full-model view because it directly asks how much of the trained model's own performance is retained.

4 Results

4.1 Backtest and allocation behaviour

Figure 2 and Figure 3 summarise the latest HTGNN checkpoint. The data split is chronological: train from 2018-01-01 to 2023-12-29, validation from 2024-01-01 to 2024-12-31, and test from 2025-01-02 to 2025-12-31. The displayed backtest uses the configured strategy-evaluation window 2024-01-01–2025-12-31, so it covers both the validation year and the test year. The accumulated-return curve separates clearly from the USD benchmark and from the hindsight best single currency, CHF. The model reaches 256.2% cumulative return and 84.6% annualized return, with 4.8% annualized volatility and −0.84% maximum drawdown. CHF, the best single-currency benchmark, reaches only 6.0% cumulative return.

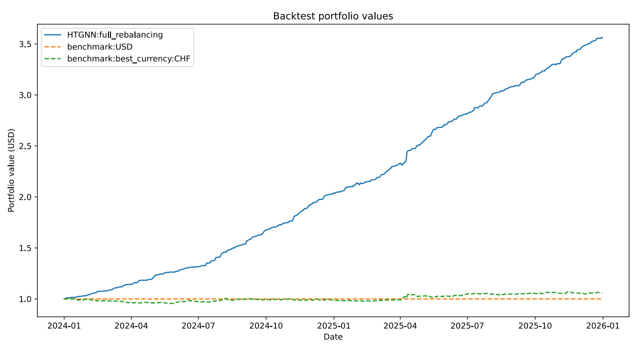

Figure 2: Accumulated returns for the latest HTGNN checkpoint against enabled benchmarks.

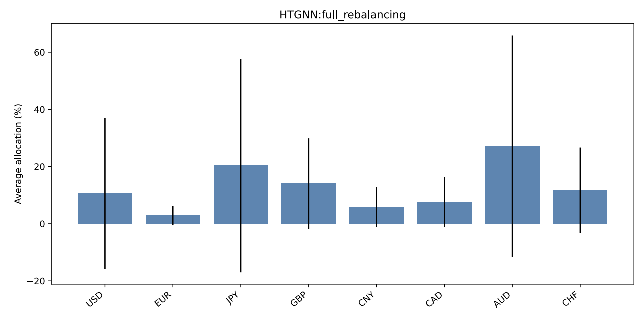

Figure 3: Average currency allocations for the latest HTGNN checkpoint.

The allocation histogram in Figure 3 is important because it rules out a trivial explanation of the return curve. The strategy does not simply hold CHF or collapse into one currency. Average allocation is largest in AUD (27.1%) and JPY (20.3%), followed by GBP (14.0%), CHF (11.7%), USD (10.5%), CAD (7.6%), CNY (5.9%), and EUR (2.8%). This supports the interpretation that the HTGNN is learning a diversified relative-value policy, although the high turnover observed in the backtest still makes transaction-cost analysis necessary.

Metric	HTGNN	USD	Best single FX (CHF)
Cumulative return	256.2%	0.0%	6.0%
Annualized return	84.6%	0.0%	2.8%
Annualized volatility	4.8%	0.0%	5.0%
Sharpe ratio	12.84	0.00	0.56
Maximum drawdown	−0.84%	0.0%	−4.65%
Mean turnover	0.703	0.000	0.000

Table 1: Backtest metrics for the latest HTGNN checkpoint and enabled benchmarks.

4.2 Key XAI results

Figure 4 shows the absolute SHAP grid for predicted mean allocations. The strongest cells concentrate in the AUD output. JPY, AUD, EUR, GBP, CAD, USD, CHF, and CNY all contribute to the AUD allocation, with JPY and AUD largest. This is useful because AUD is also the largest average allocation in Figure 3. The signed version in Figure 8 is almost zero everywhere. The contrast is informative: absolute SHAP reveals that the portfolio-signal inputs matter in magnitude, while signed SHAP shows that their direction is not stable across samples and largely cancels out. However, the SHAP surrogate has low fidelity: the mean-allocation surrogate R^2 is 0.050. The correct conclusion is therefore not that the whole HTGNN is linear in the portfolio signal, but that within this restricted portfolio-signal view, AUD is the most sensitive output.

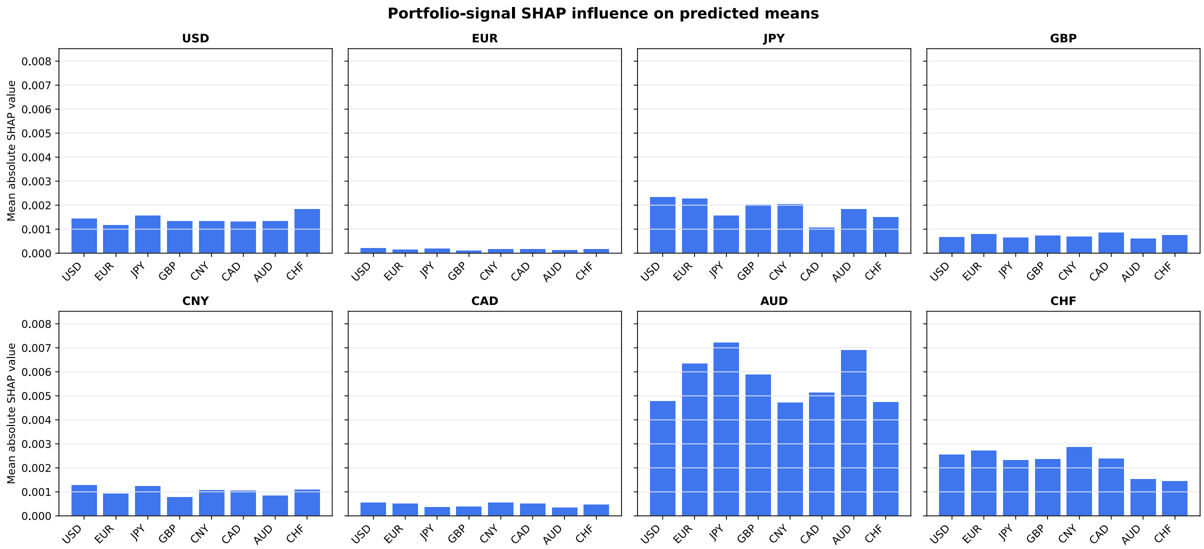

Figure 4: Absolute portfolio-signal SHAP influence on predicted mean allocations. Each subplot fixes one output currency.

Integrated Gradients, shown in Figure 5, gives a broader full-graph view. Five nodes receive especially high focus: `fx_future`, `commodity_future`, `portfolio_signal`, `fx_usd_pair`, and `commodity_etf`. This supports a market interpretation of the HTGNN: direct currency derivative information is central, commodities help contextualise commodity-sensitive currencies such as AUD and CAD, and the portfolio signal still matters as a rolling allocation state. The histogram in Figure 11 shows the same ranking and, importantly, small variance intervals for each node's focus, meaning that the local examples do not produce a highly unstable attention pattern.

Given the strong backtest, one possible concern is data leakage: perhaps one feature inside a highly attended node is accidentally carrying future information. The feature-level pie charts in Figure 13 reduce that concern: inside the most attended nodes, attribution is visually spread across several features rather than concentrated in one suspicious channel. This does not prove absence of leakage, but it makes a single-feature leakage explanation less plausible.

Finally, the generic latent nodes are almost zero in Figure 5 because that IG graph propagates gradients all the way to observed input features, and the generic nodes are not observed input-feature blocks. The last-message graph in Figure 12 stops the attribution at the states before the final message passing layer. It shows that generic nodes are not exactly zero, but their focus remains negligible, around 0.08–0.09% each, compared with `portfolio_signal`, `fx_future`, and `fx_usd_pair`. This suggests that the model is not making effective use of the generic nodes to model useful cross-feature information across the observed feature nodes.

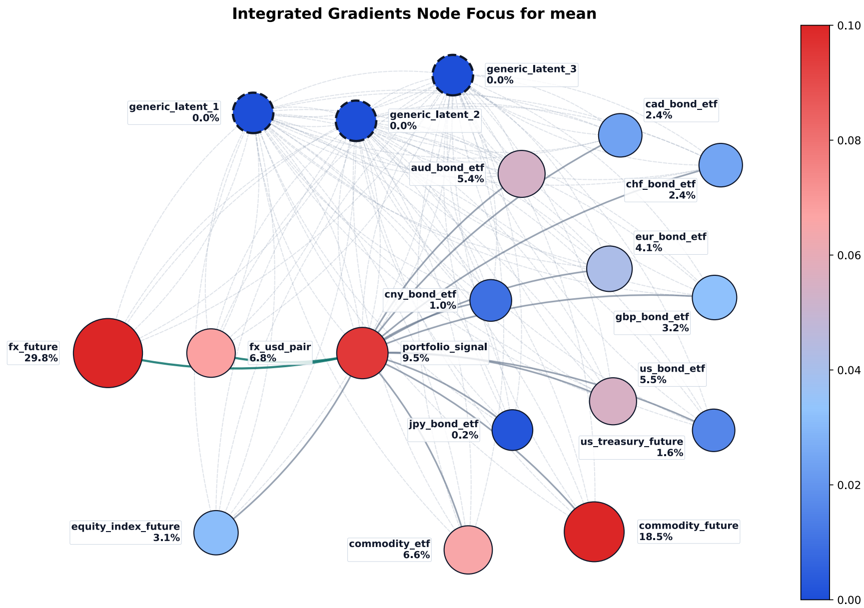

Figure 5: Integrated Gradients node focus on the HTGNN topology.

Figure 6 and Figure 7 combine two perturbation results. Deletion/insertion relative to the full model indicates that the first three removed nodes (`fx_usd_pair`, `equity_index_future`, and `portfolio_signal`) only modestly reduce retained performance. The major break appears when `fx_future` is also deleted: retained performance falls to roughly 65% of the full model. Insertion mirrors this behaviour: restoring the first three nodes is not enough, while restoring the fourth recovers almost full-model performance. This suggests joint dependence and partial redundancy among the top market blocks.

The temporal occlusion panel in Figure 7 shows that the most recent window, days 16–20, is the most helpful: masking it lowers the performance ratio by about 0.118. By contrast, masking days 12–16 improves the performance ratio by about 0.103. This is one of the most actionable XAI findings: the model benefits from very recent information but may also carry a mid-window pattern that is harmful on the test sample.

The node-wise temporal occlusion appendix figure (Figure 14) adds a negative nuance for `commodity_future`. This node receives high IG focus, but occluding it does not produce a robust performance drop across windows: the perturbed strategy remains profitable in every window, and for several later windows the drop is negative, meaning that masking the node actually improves the performance ratio. This suggests that commodity futures are influential but may inject noisy or harmful temporal context in part of the test period.

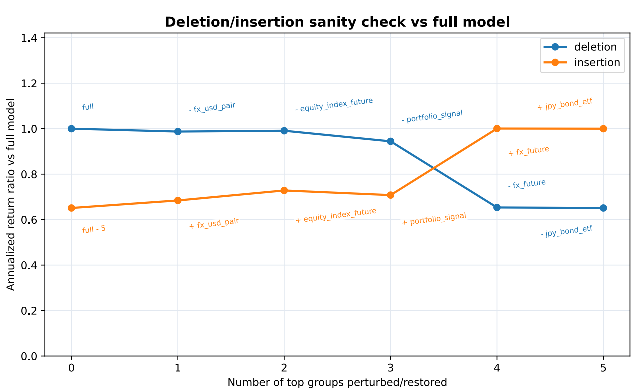

Figure 6: Deletion/insertion faithfulness check relative to the full model.

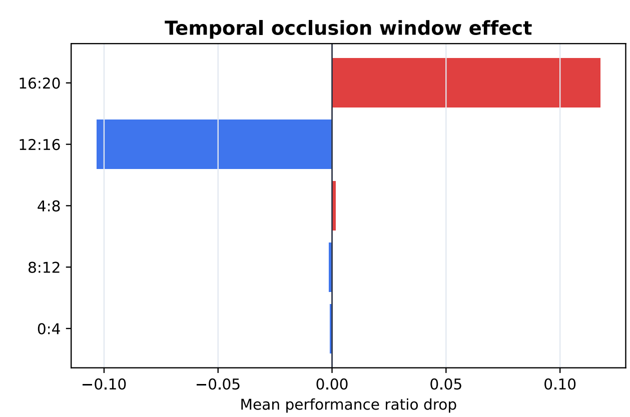

Figure 7: Global temporal occlusion over five lookback windows.

Additional figures are included in the appendix: signed SHAP for mean, absolute/signed SHAP for variance, the IG node histogram, the last-message IG graph, IG feature pie charts, and node-wise temporal occlusion. The deletion/insertion curve relative to the equal-weight benchmark is omitted because Figure 6 is more directly aligned with model faithfulness.

5 Actions and Insights

5.1 Insights from XAI

The XAI results point to a coherent model story. First, the strongest global allocation behaviour is AUD-heavy, and the portfolio-signal SHAP grid also shows AUD as the output most sensitive to recent allocation-regime inputs. Second, the full-graph IG view shifts attention from currencies to market channels: `fx_future`, commodities, and the portfolio signal dominate. Third, the perturbation checks show that the model is not dependent on one isolated node; performance degrades sharply only after several top nodes are removed.

Taken together, these observations suggest that the HTGNN uses both direct FX information and broader macro-financial context. This is exactly the type of behaviour the heterogeneous graph architecture was designed to allow. The explanations also identify a weaker point: the model is temporally asymmetric. Recent information is useful, but the previous four-day window can be harmful. That finding is not visible from backtest performance alone.

5.2 Model actions

The first action is to improve temporal selectivity. Since the most recent window helps and the 12–16 window hurts, the model should be extended with a learned temporal attention or gating mechanism before message passing. This would let the model learn which parts of the 20-day sequence should be emphasised rather than compressing the whole window through a GRU state.

The second action is explanation monitoring. The IG graph indicates that `fx_future` and commodity futures are major drivers. In a future deployment, data quality alerts should be node-specific: stale FX futures or missing commodity data should be treated as model-risk events, not merely as generic missing values.

The third action is not to over-correct every surprising XAI result. Low signed SHAP values are not necessarily a problem because signed contributions cancel by construction when effects change sign across samples. Similarly, redundancy in deletion/insertion is not automatically bad; it can indicate that the graph has multiple related channels carrying overlapping market information. What should be mitigated is unexplained instability, not every non-sparse explanation.

5.3 Domain recommendations

For portfolio use, the model should be reported through economic channels: "FX derivatives", "commodity-linked risk", "portfolio-regime signal", and "equity-risk context", rather than raw tensor names. This makes the explanation usable for risk committees and clients.

The model should also be evaluated under transaction costs and turnover constraints before any realistic trading interpretation. The backtest return is large, but mean turnover is 0.703, so implementation frictions could materially change performance. Finally, because FX regimes are policy-sensitive, any allocation recommendation should be accompanied by rolling-window and stress-period explanations. A model that looks interpretable in a calm period may rely on different nodes during a central-bank or geopolitical shock.

6 Discussion: Limitations, Risks, and Future Work

The main limitation is the short evaluation horizon. The 2024–2025 backtest is strong, but FX markets can change abruptly through inflation surprises, monetary-policy divergence, interventions, and geopolitical events. A two-year test period is not enough to establish structural profitability.

The second limitation is explanatory fidelity. SHAP is deliberately restricted to `portfolio_signal`, and its ridge surrogate has low R^2. It is useful for a focused question about allocation-signal currencies, but not a full explanation of the HTGNN. IG and perturbation methods cover the full graph, but they depend on the background mean baseline. A different baseline, such as zero, rolling normal conditions, or stress-period averages, could change the magnitude of the explanations.

There are also model-risk and ethical risks. A high-return black-box allocation model can encourage automation bias: users may trust the output because the backtest is strong, even when the explanation says the model depends on a small set of market blocks. The model may also transfer poorly to regimes where central-bank interventions dominate historical relationships. Finally, the strategy may create operational risk through turnover, liquidity constraints, and unmodelled transaction costs.

Future work should include rolling retraining, transaction-cost-aware training, crisis-period slicing, explanation stability metrics, and relation-level XAI. The deletion/insertion result especially motivates interaction-aware explanations: the model appears to depend on groups of nodes jointly, so future work should measure node-pair and relation-type effects, not only individual node scores.

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Appendix: Additional XAI Figures

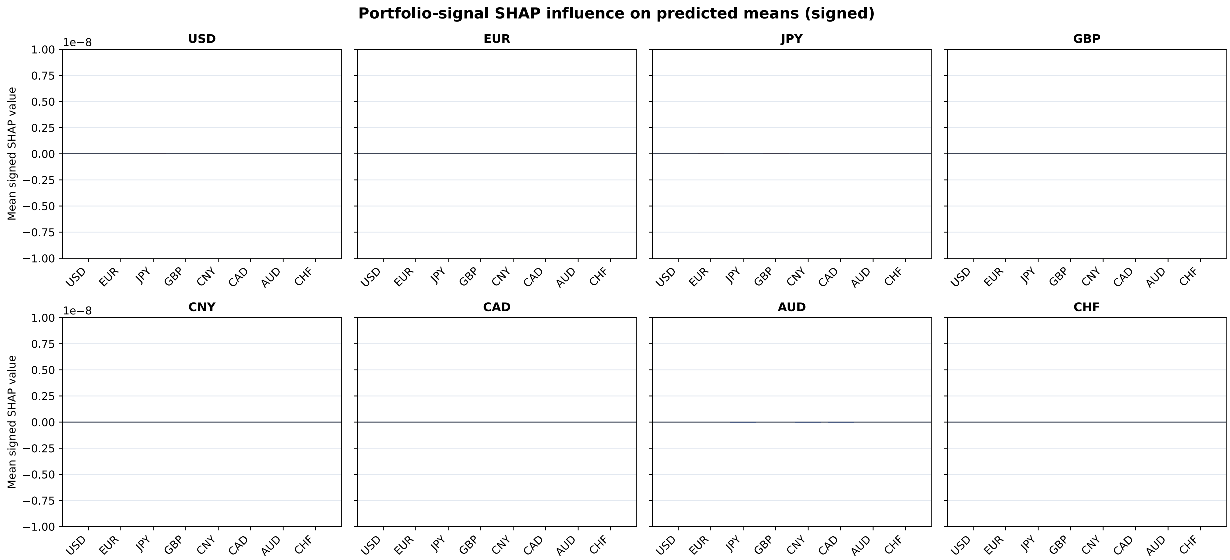

Figure 8: Signed SHAP influence on predicted mean allocations. This view is useful for net direction, but cancellations make magnitudes small.

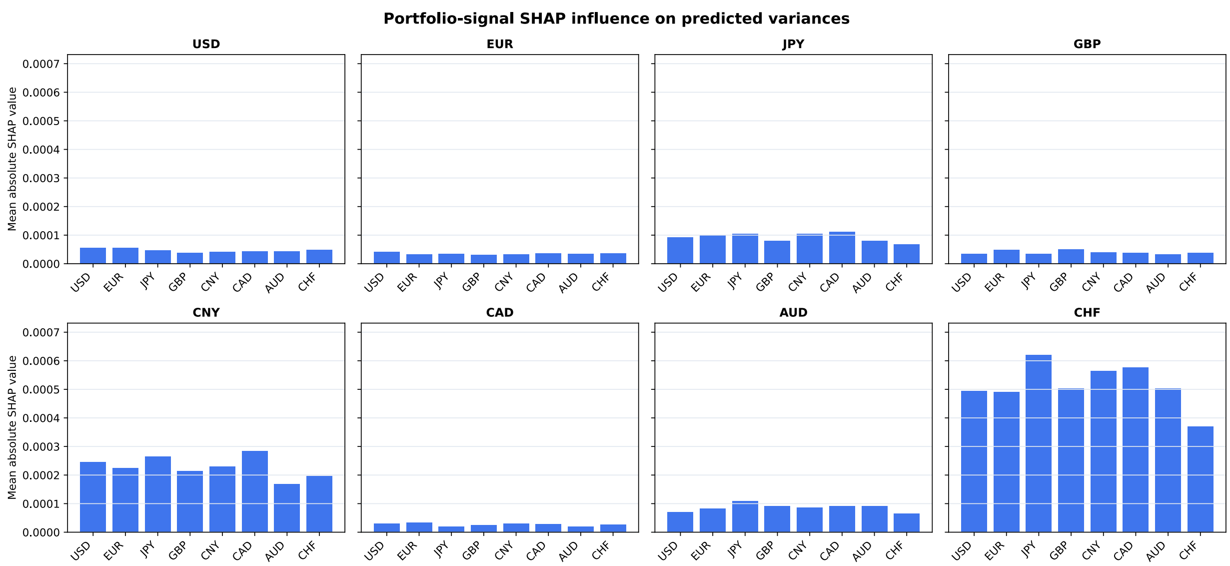

Figure 9: Absolute SHAP influence on predicted marginal allocation uncertainty.

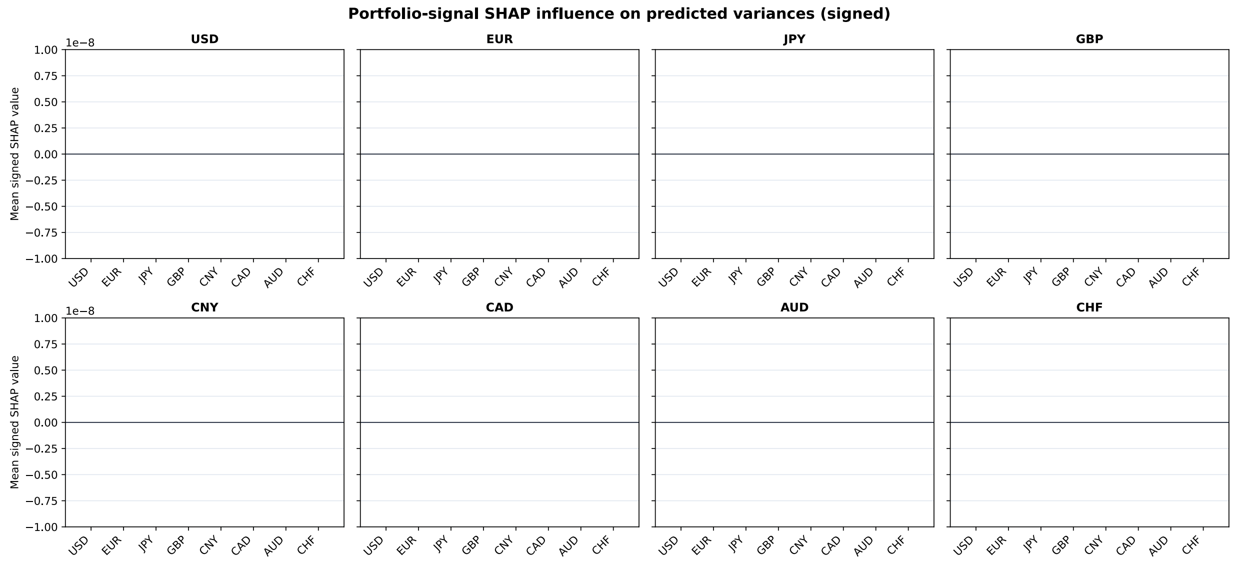

Figure 10: Signed SHAP influence on predicted marginal allocation uncertainty.

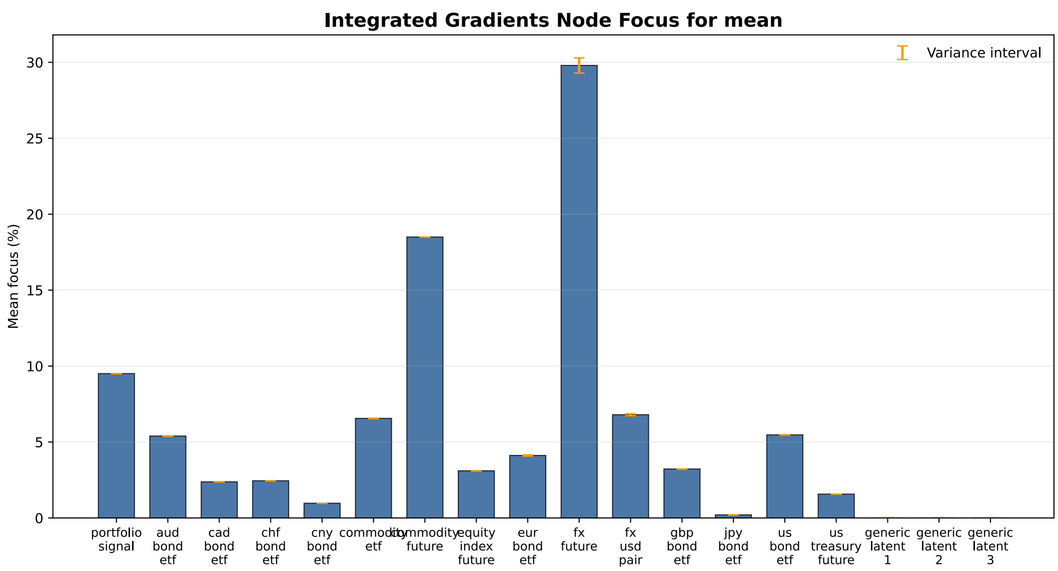

Figure 11: Integrated Gradients node-importance histogram.

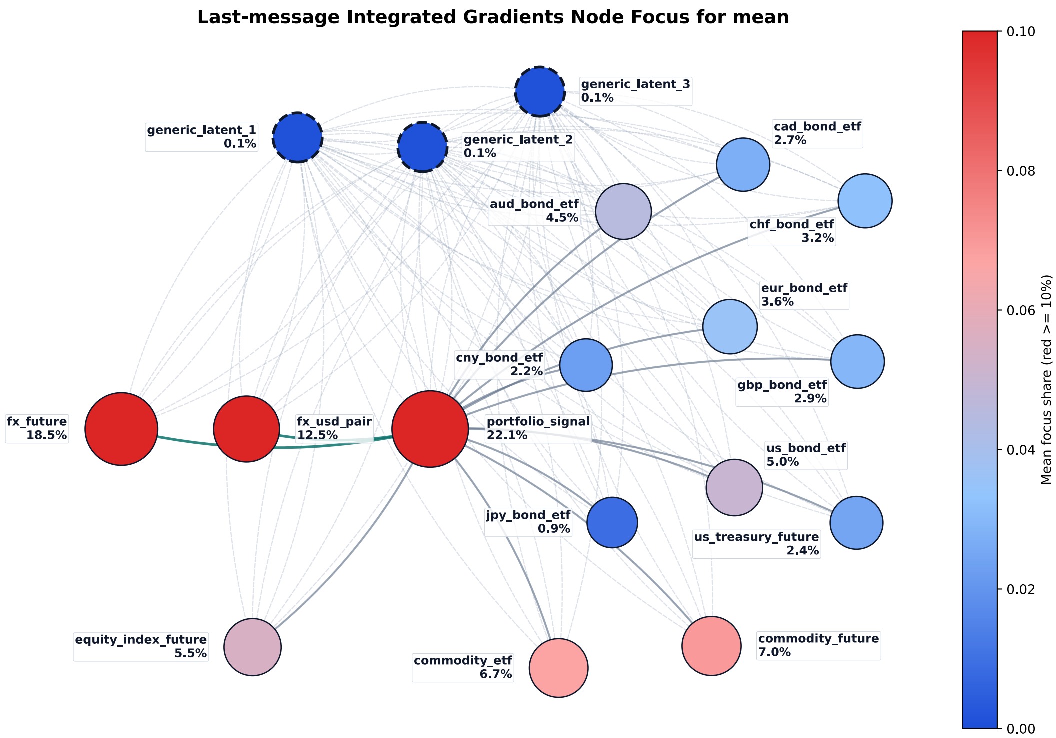

Figure 12: Integrated Gradients node focus when attribution starts before the final message passing layer.

Figure 13: Feature-level decomposition inside each observed node using Integrated Gradients.

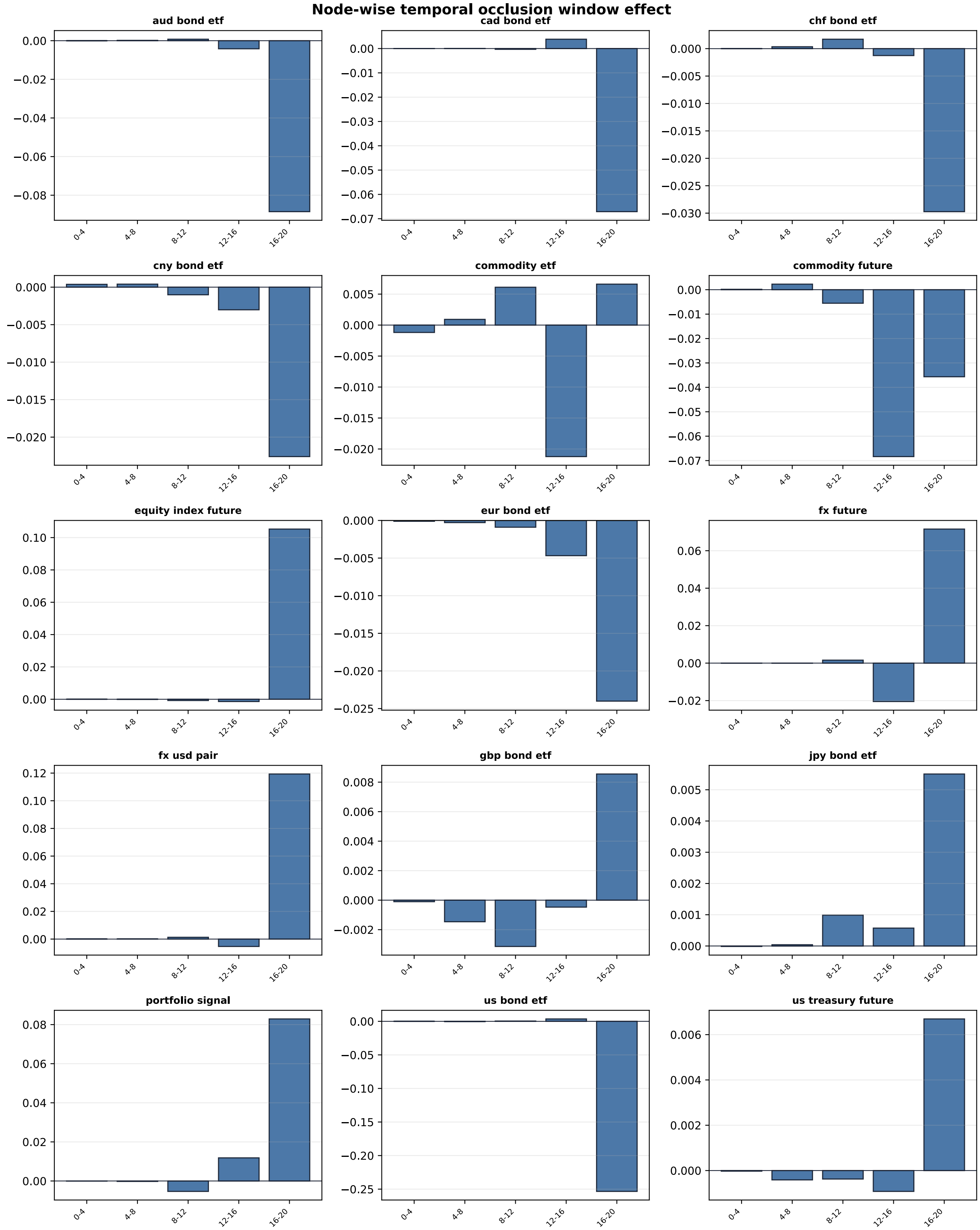

Figure 14: Node-wise temporal occlusion: five vertical mini-histograms per node.